



SRv6 Decentralized SDN (DSDN)

Architecture Overview & Roadmap Update

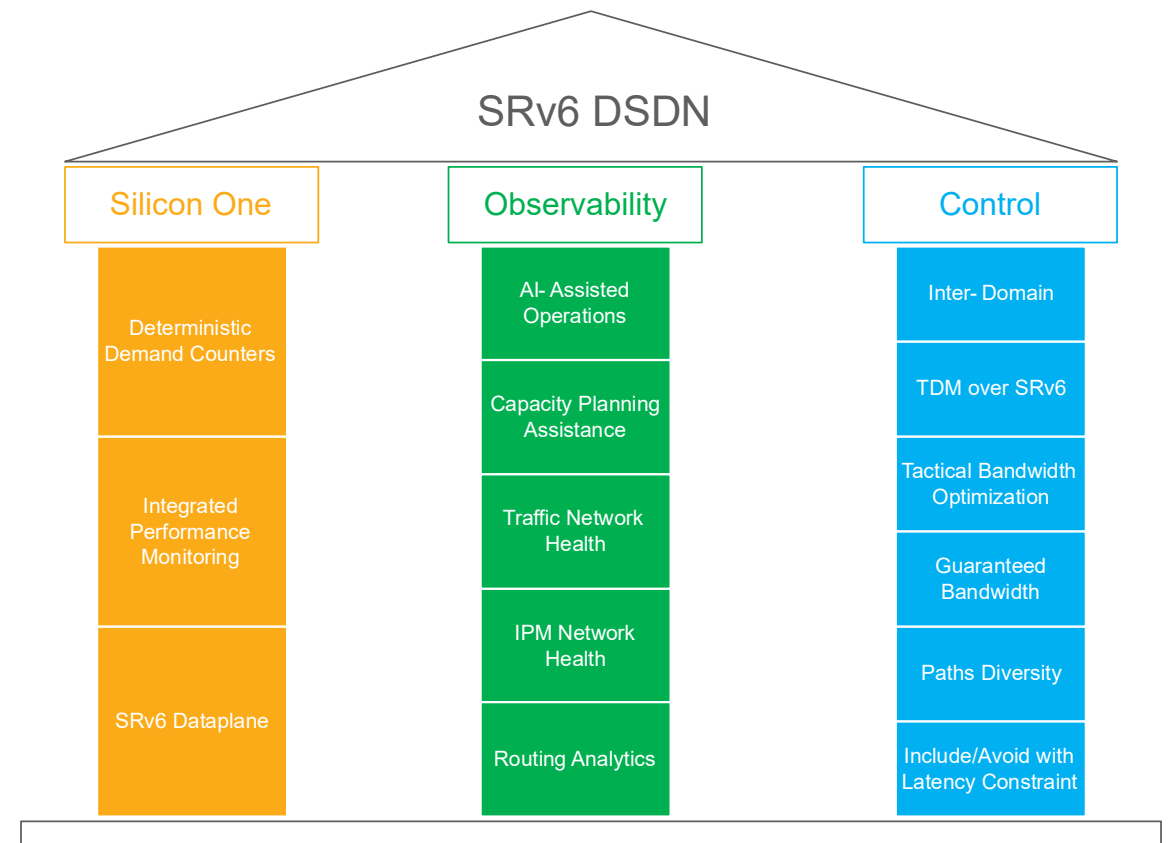
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Three-Pillar Architecture

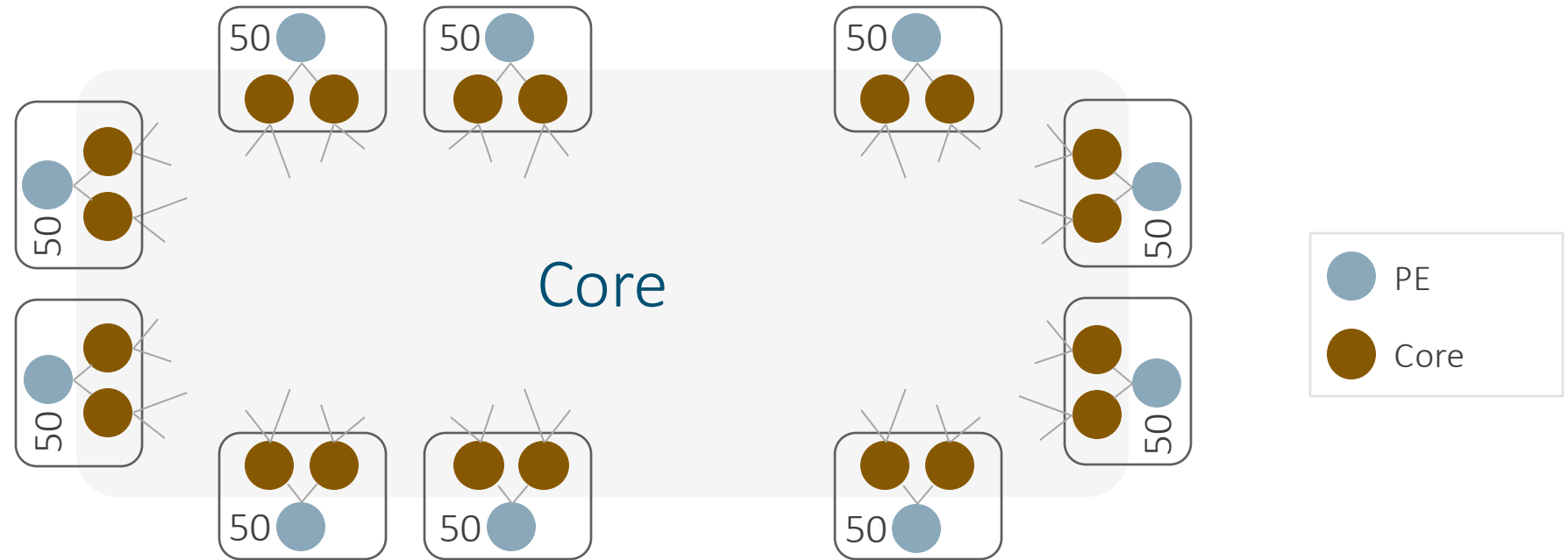
- Silicon One **enables** everything
- Observability **informs** decisions
- Control **acts** on that intelligence



Observability

Integrated Performance Measurement (IPM)

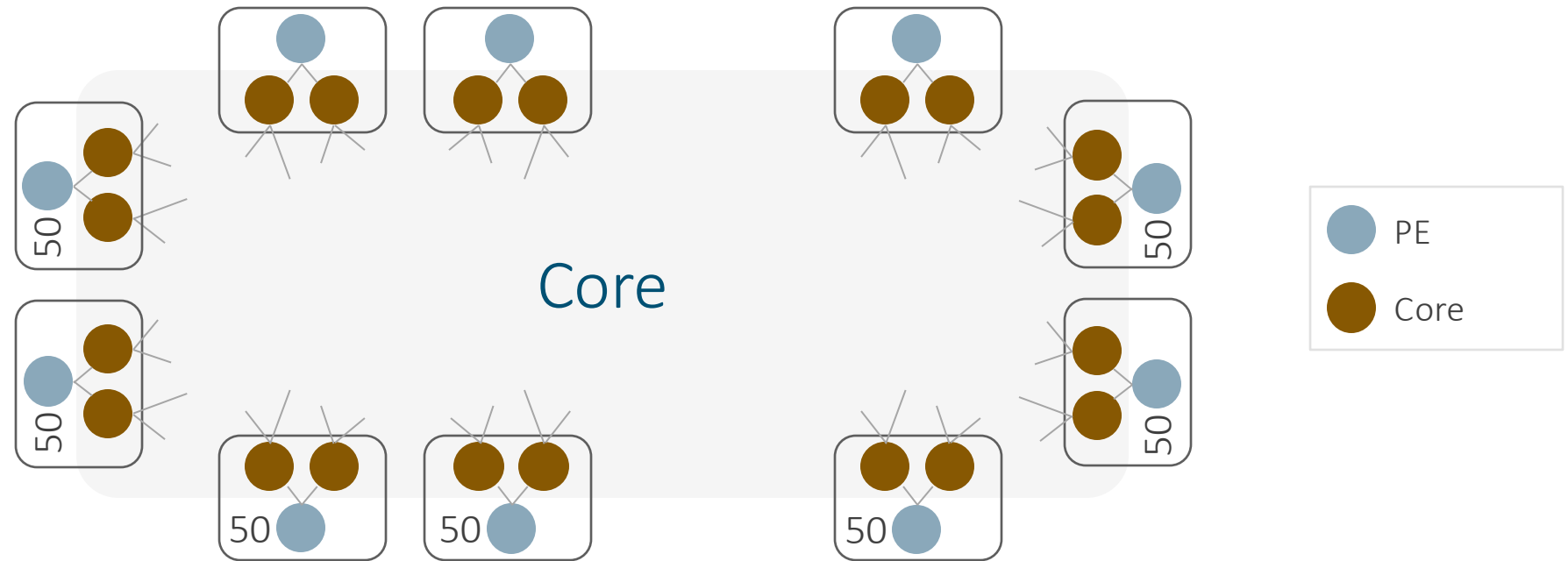
The nature of IP is ECMP



- 8 ECMP paths between each pair of PEs
 - 800 PEs
- $800 \times 8 = 6.4\text{k}$ paths from each PE to all other PEs

How many probes ?

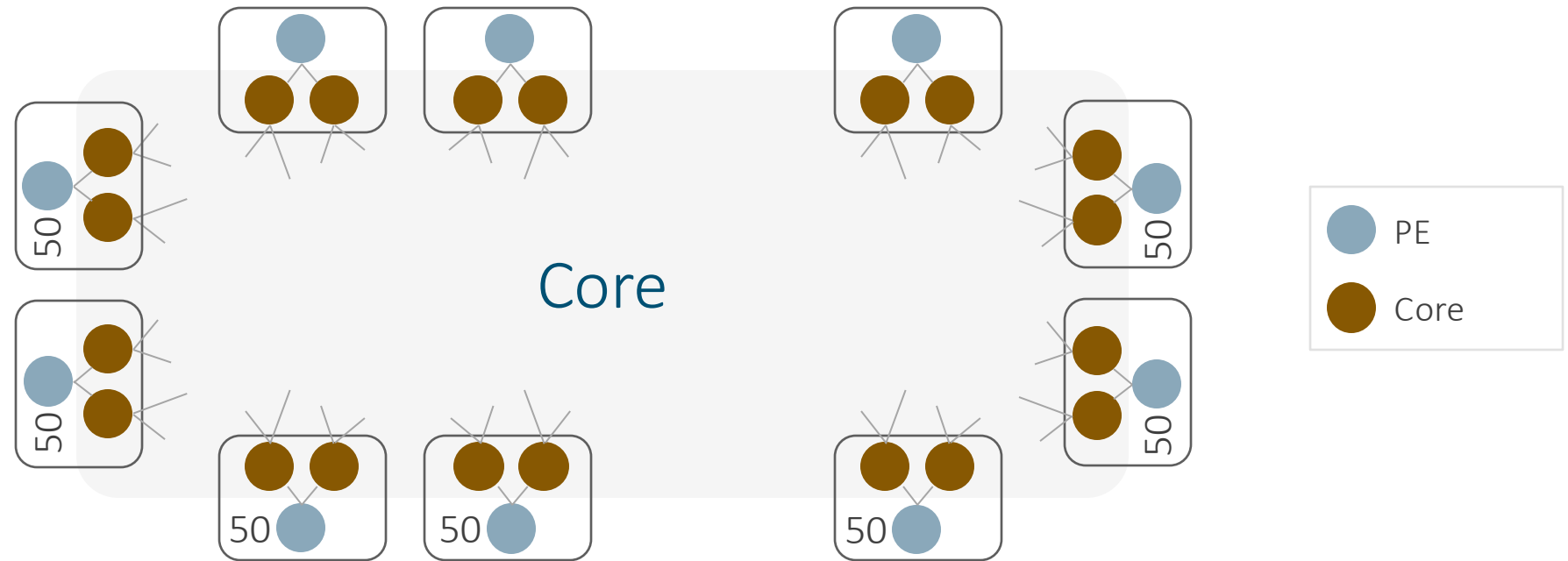
- Measure **all ECMP paths** between **all PE pairs** at **millisecond** granularity:



- 1 measurement every msec
 - 800 PEs
 - 8 ECMP paths
- 6.4M probes per sec

How many probes ?

- Measure **all ECMP paths** between **all PE pairs** at **millisecond** granularity:

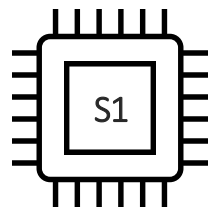


- 1 measurement every msec
- 800 PEs
- 8 ECMP paths

6.4M probes per sec

(45% of Silicon One capability)

14 Million Probes Per Second



1000 to 10000 times better scale than legacy solutions



Legacy solutions are typically 1000 to 10000 times not scalable enough

Legacy coverage is $< 0.1\%$

Operators learn outages from clients

Would a bank accept to monitor $< 0.1\%$ of its access?

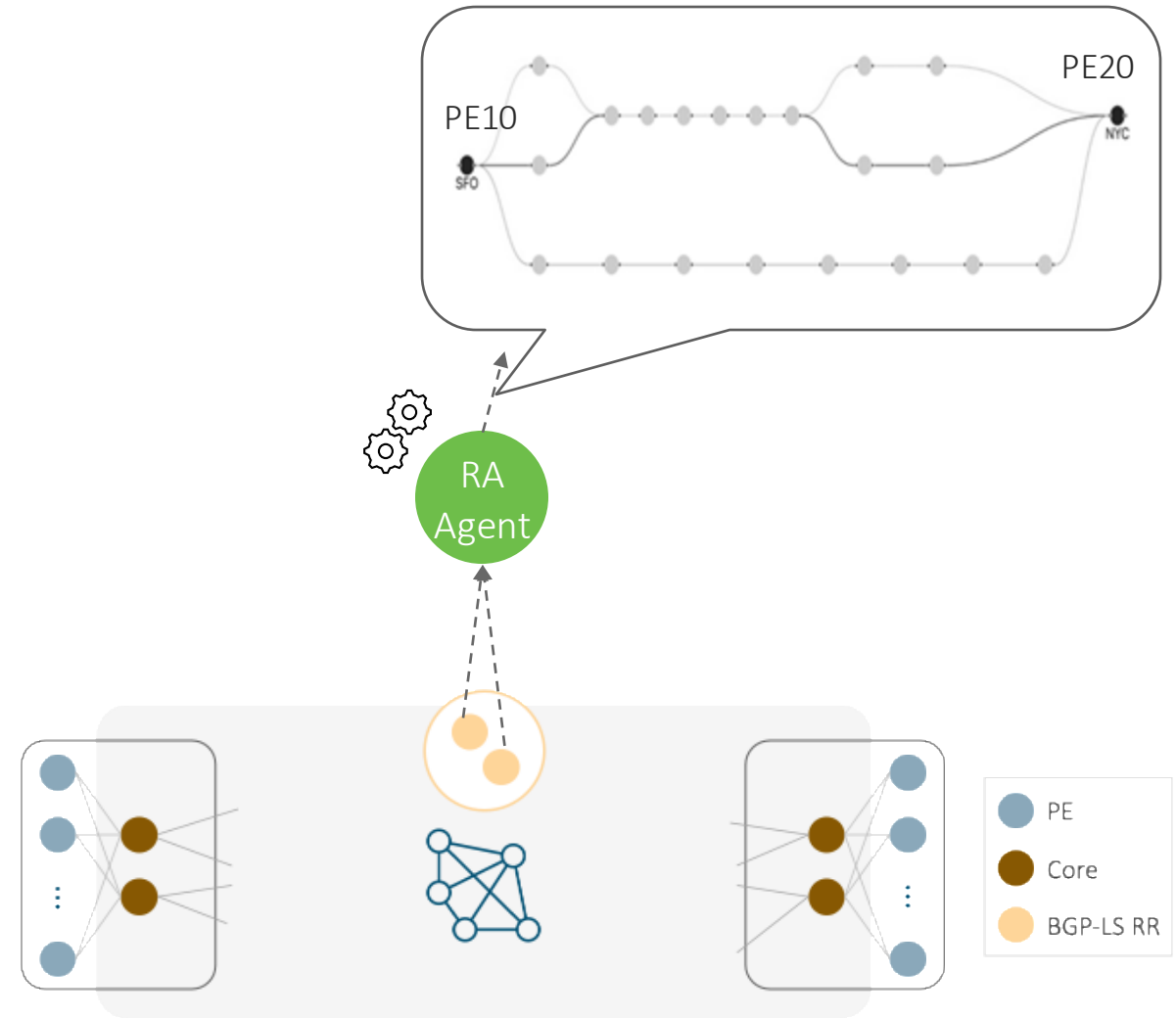
IPM

- Scale: 14M PPS out & 14M PPS in
- Latency: One-way & histogram
- Absolute loss: alternate marking (RFC 9341)
- Liveness detection in HW (sub-2msec)
- Standard-based (RFC 8762, RFC 8972)
- Much cheaper - capex and opex elimination
- Deterministic per-ECMP path measurements
 - The power of IPM + SRv6

Routing Analytics

Routing Analytics (RA) Agent

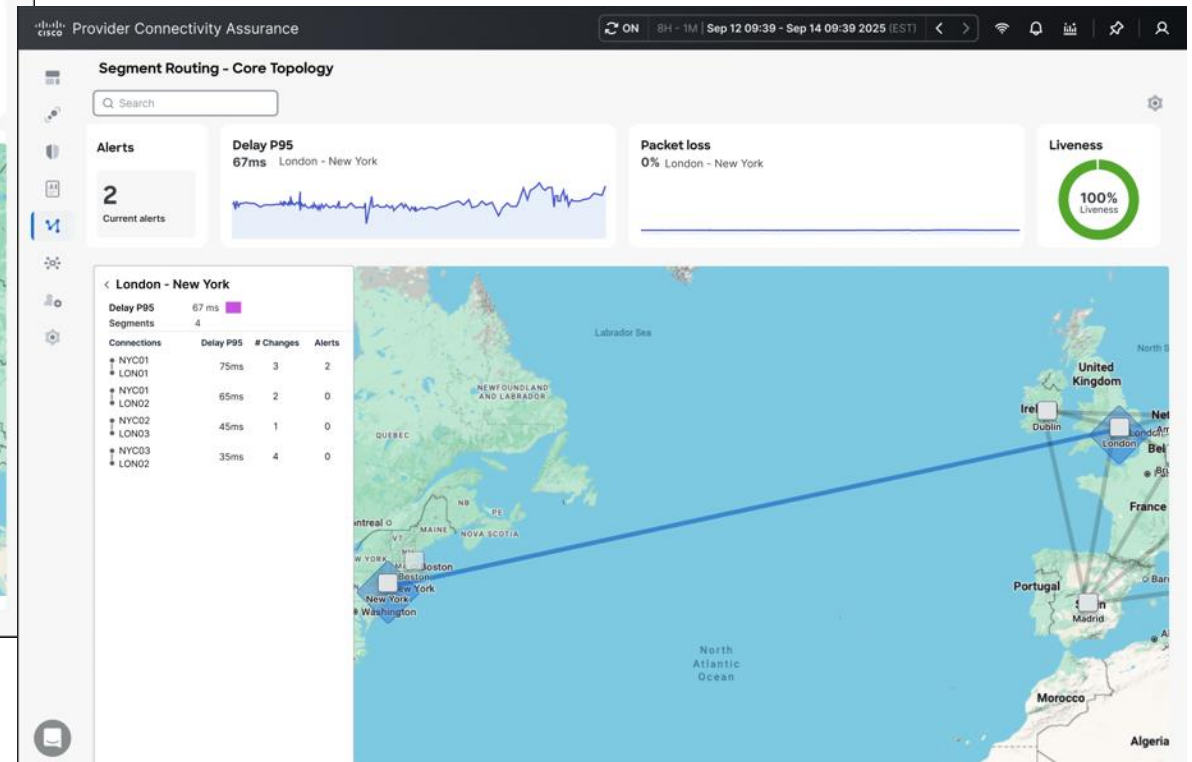
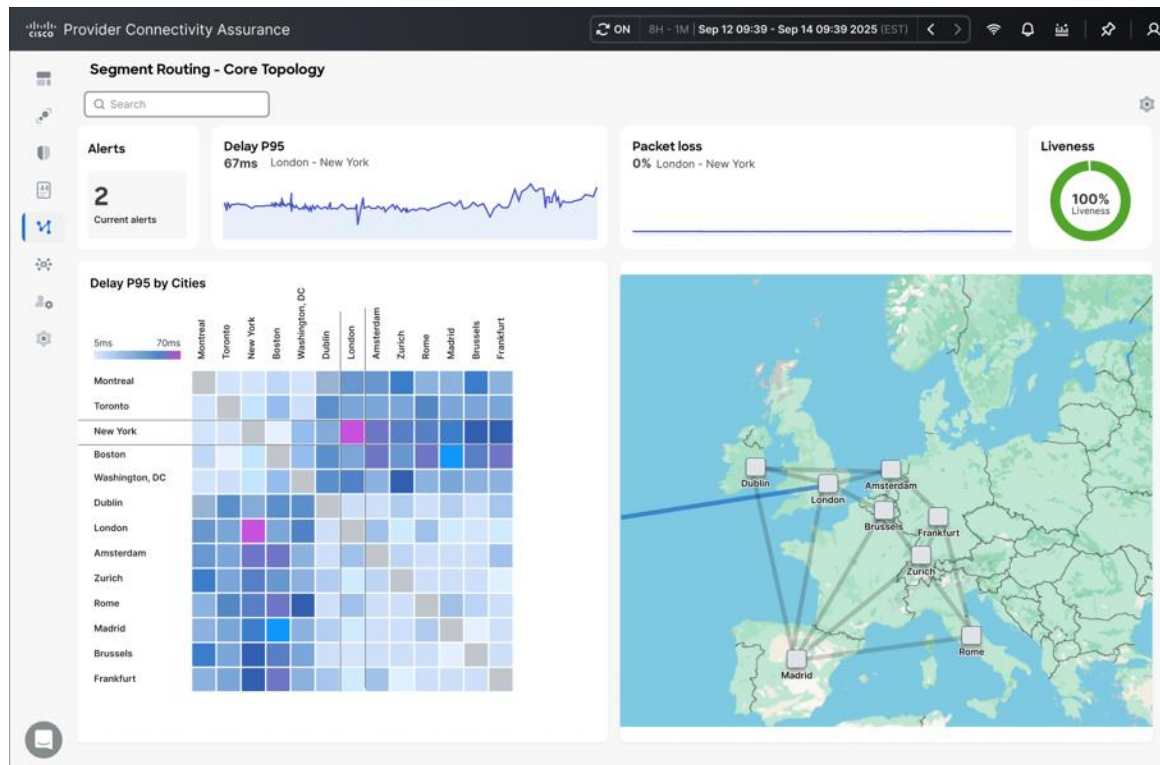
- Monitor the **routing topology** in **real-time** using **BGP-LS**
- Compute **each path**
 - All Flex-Algos and SR Policies
 - **Any PE to any PE**
- Report every time any path changes:
 - New ECMP path set
 - End-to-end propagation delay for each path



Crosswork Assurance

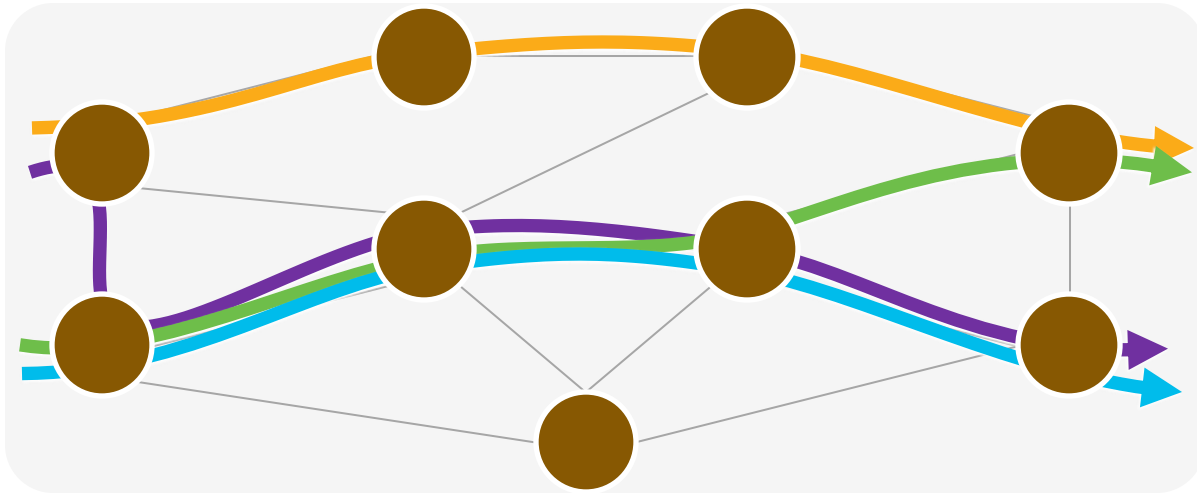
Crosswork Assurance (CA)

- CA ingests and keeps timeseries of the IPM and RA data
- CA correlates the data and presents the outcome in a rich dashboard



Deterministic Demand Matrix (DDM)

Demand Matrix: A Decades-Old Problem



from \ to				
	D1	D2	D3	D4
S1	—	?	?	?
S2	?	—	?	?
S3	?	?	—	?
S4	?	?	?	—

- **Automated**: Not everyone can afford a dedicated team to acquire the demand matrix.
- **Real-time**: Reflect the current state of the network.
- **Deterministic**: Show all the traffic. No sampling or approximations
- **Lightweight**: No heavy collection or processing infrastructure.

SRv6 Deterministic Demand Counters

SRv6 Locator counters

- [Per-locator, per-Algo](#) granularity
- Native support for summarization

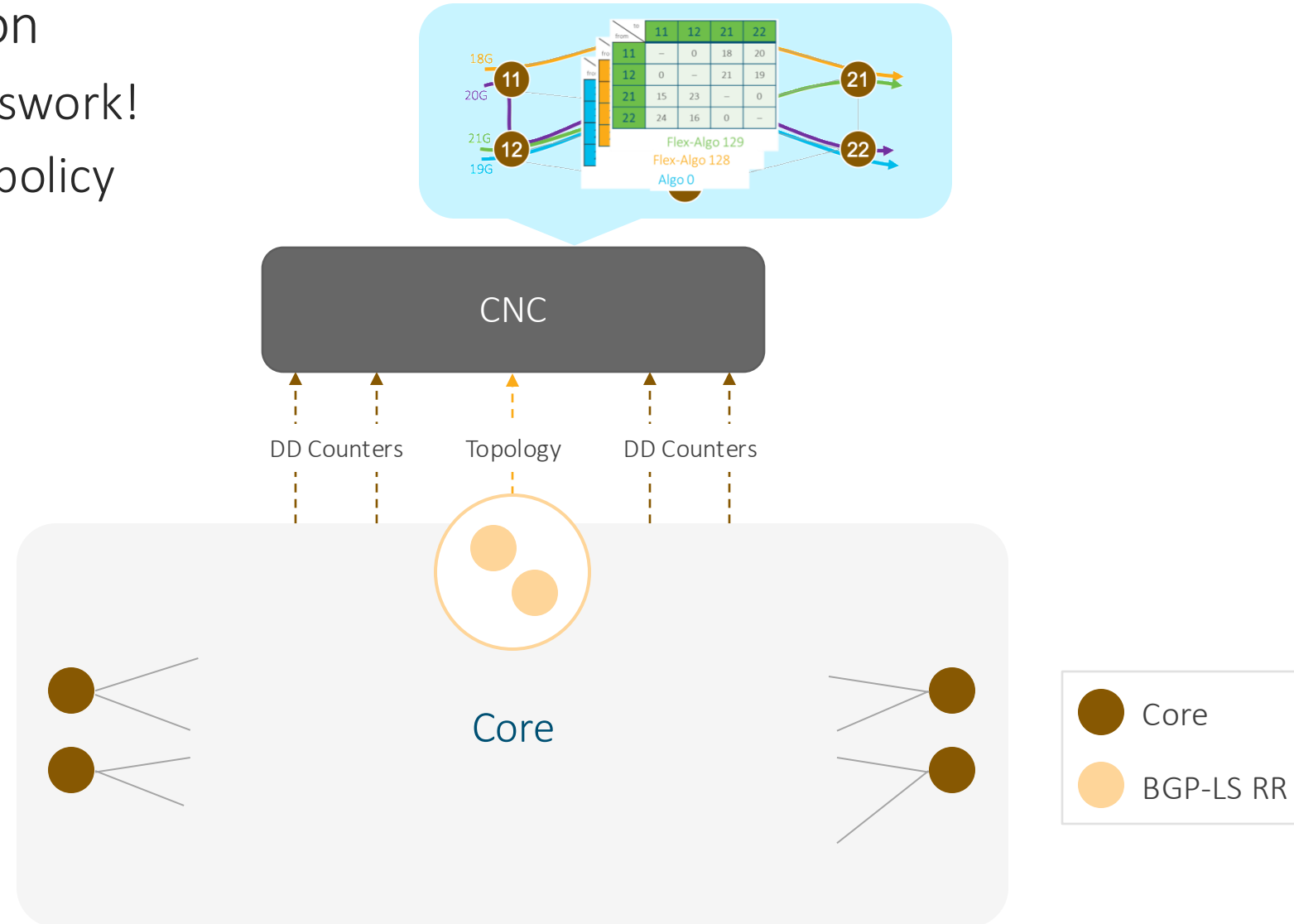
SRv6 Policy counters

- [Per-segment-list](#) granularity

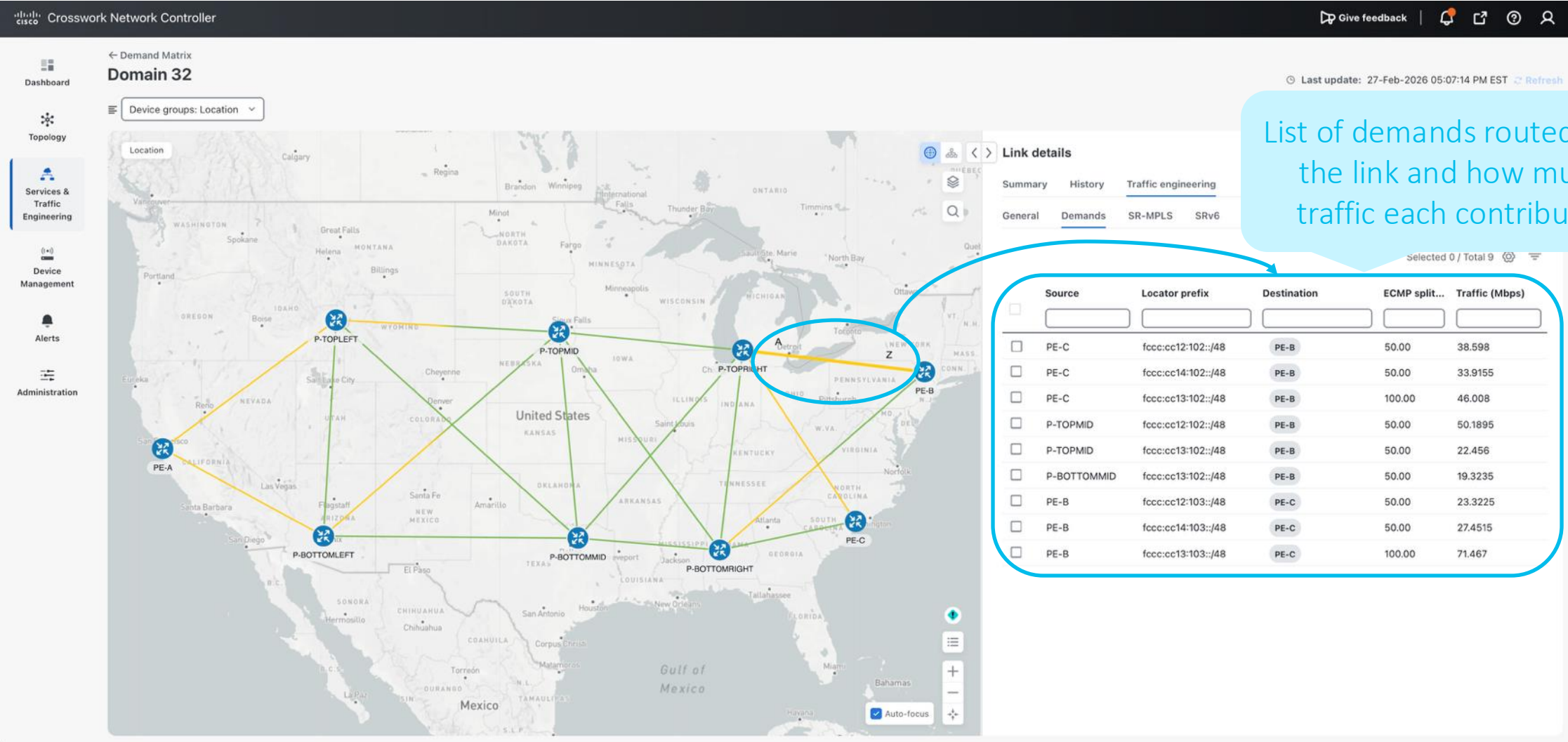
- Accounting in [hardware](#) for maximum precision
- [Telemetry export](#) via gNMI or YANG-Push
- Supported on [Silicon One](#) and [across the SRv6 portfolio](#)
(c8000, NCS 5500 and 5700 families, ASR 9000)
- Ongoing IETF [standardization](#) (draft-ietf-spring-srv6-yang, draft-ietf-spring-sr-policy-yang)

Building the Deterministic Demand Matrix (DDM)

- Fully **deterministic** calculation
 - No approximation nor guesswork!
 - For every Flex-Algo and SR policy
- Available since CNC 7.2



DDM — What's going through this link?



Foundational Building Block

		to			
		11	12	21	22
from	11	—	0	18	20
	12	0	—	21	19
	21	15	23	—	0
	22	24	16	0	—

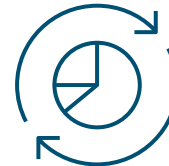
Deterministic Demand Matrix



Automated Capacity Planning

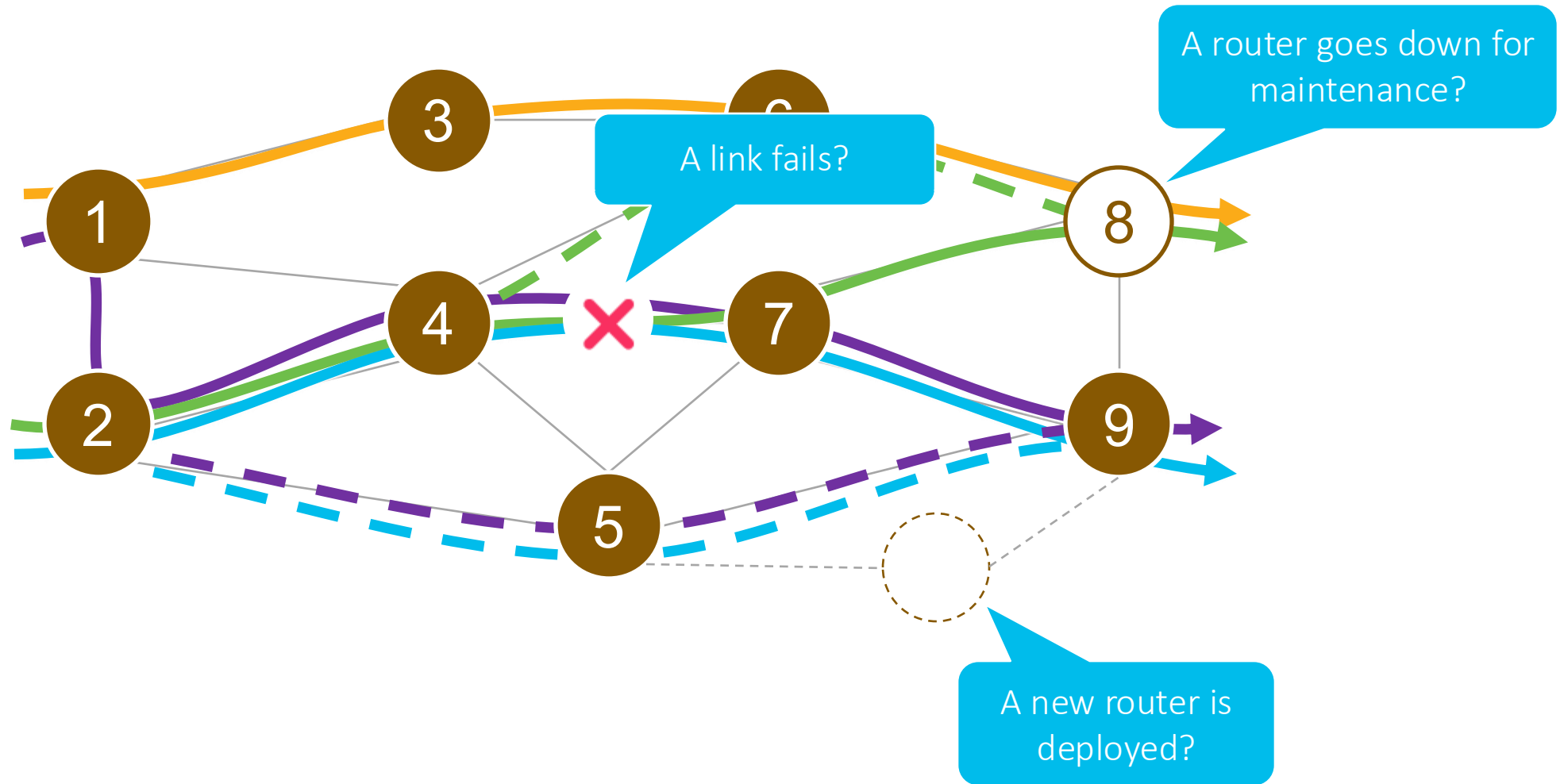


Planned Maintenance



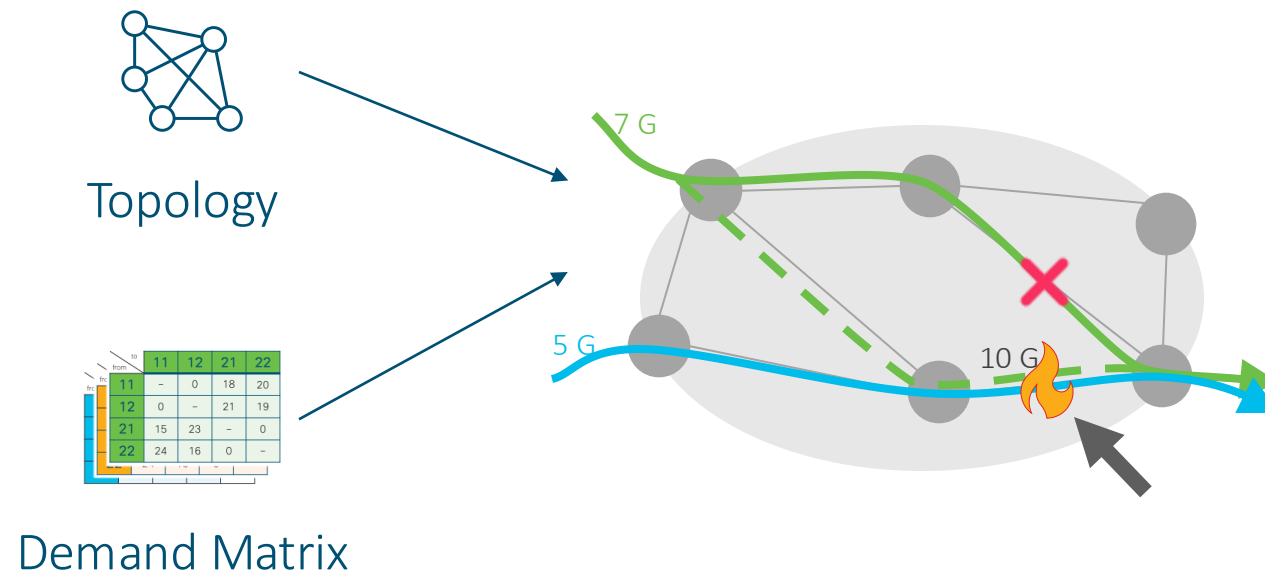
And much more...

What if...

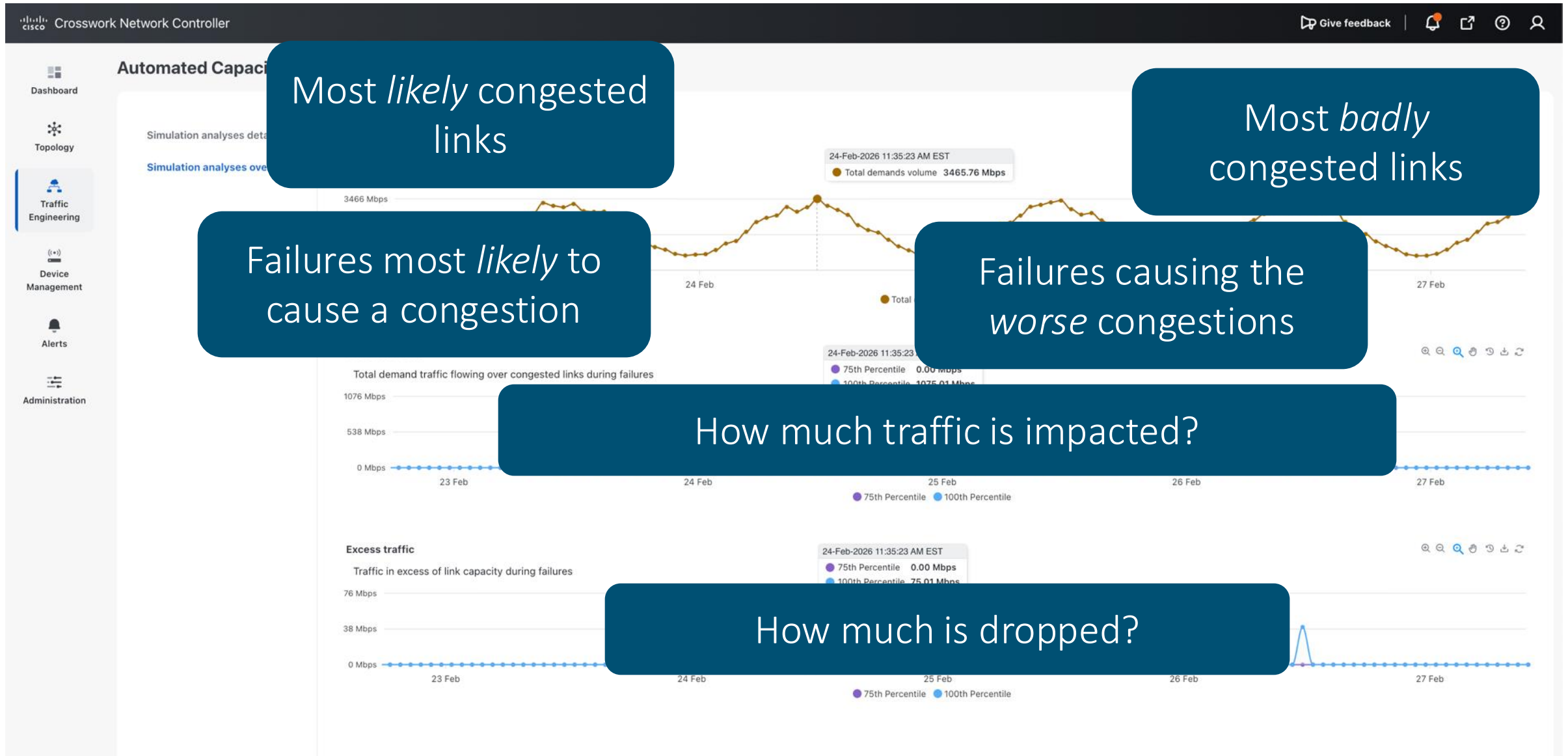


Automated Capacity Planning (ACP)

- Obtain latest **routing topology** and **demand matrix**
 - Optionally, increases demand matrix volumes by a **growth factor** (e.g., +50%)
- Simulate every single link, node, and SRLG failure scenario
- **Repeat** every 15 minutes, with daily, weekly, monthly **summaries**
- Well-thought **KPIs** make it very easy for human operator to detect risks and hot spots



CNC – ACP overview page



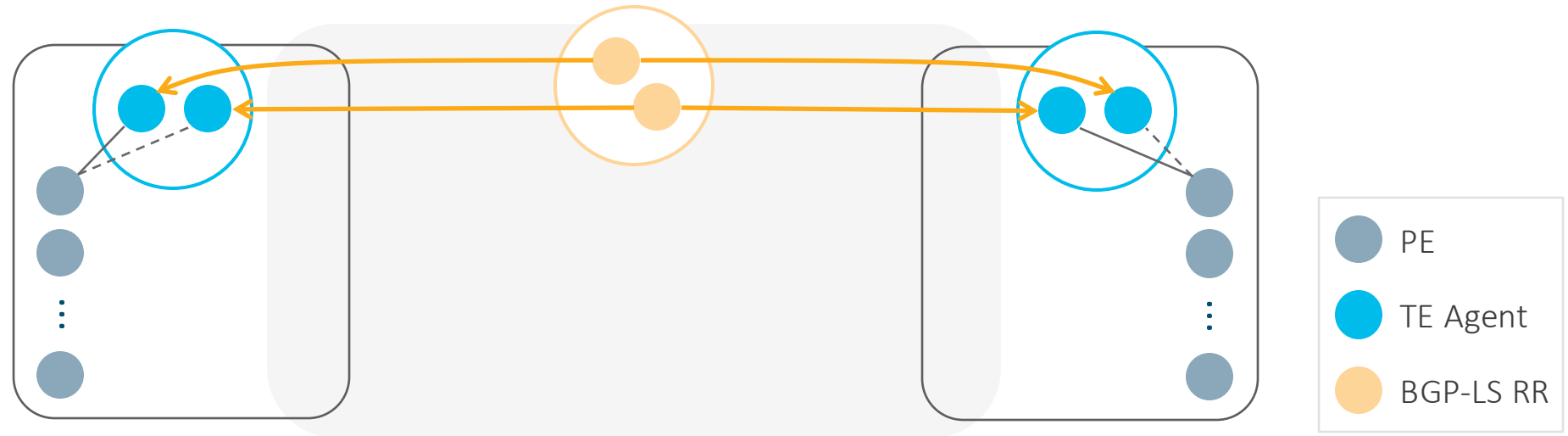
Decentralized Control

TE Agent — A Decentralized SDN Controller



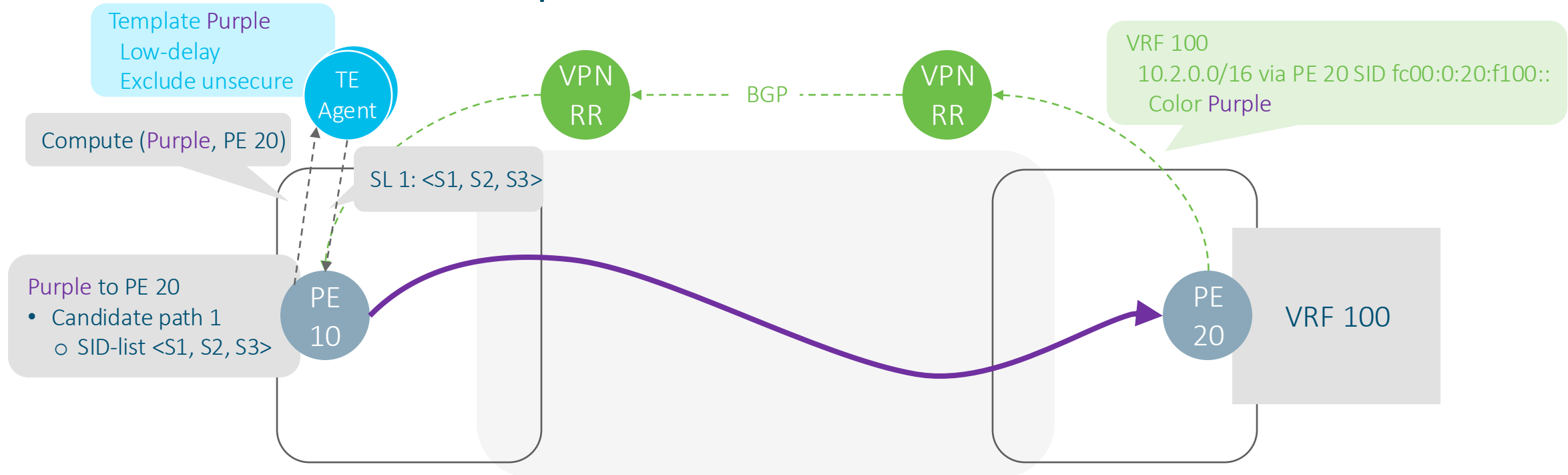
- Lightweight, Linux-based software
- Horizontally scalable like BGP RRs (e.g., 2 TE Agents per PoP, serving 50 PEs)
- Operate independently

Consensus-Free By Default



- TE Agent acquires a complete routing topology through BGP-LS
- Computes *most* SR policy paths in a **consensus-free** manner (min metric, include/exclude, delay bound, etc.)

On-Demand Next-Hop

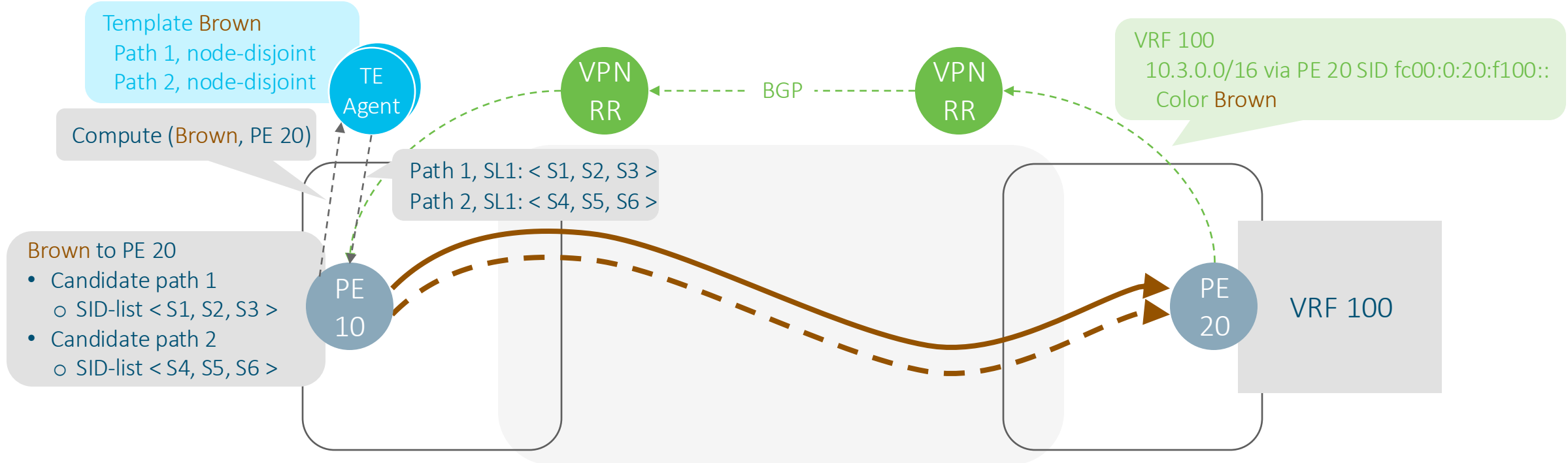


- Ingress PE receives a **colored** BGP route
- Ingress PE sends a **compute request** to TE Agent with an **endpoint** and **color**
 - Only TE agents know the color $\leftrightarrow$ intent mapping
- Ingress PE receives the SID list from TE Agent and installs it in forwarding
- Ingress PE steers traffic based on **endpoint** and **color**

Template Any Intent

- Minimize any metric
- Exclude any resource
(e.g., affinity, SRLG, IP address)
- Hop through waypoints
- Enforce disjoint paths
- Reserve bandwidth
- ... Supporting new intents
 - Only TE Agent upgrade!
 - No change to the headend

Even for Multiple Candidate Paths

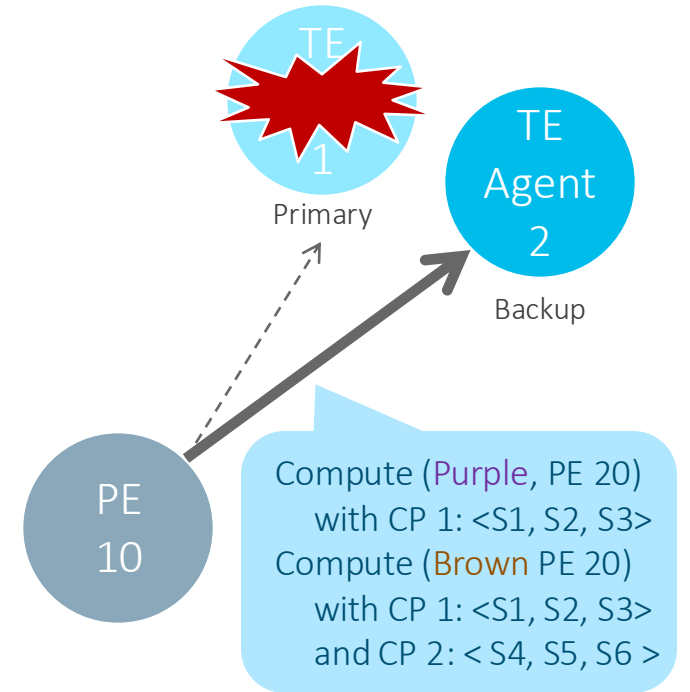


- A single compute request (**Brown**, PE 20) can result in **multiple candidate paths**

A Simple High-Availability Model

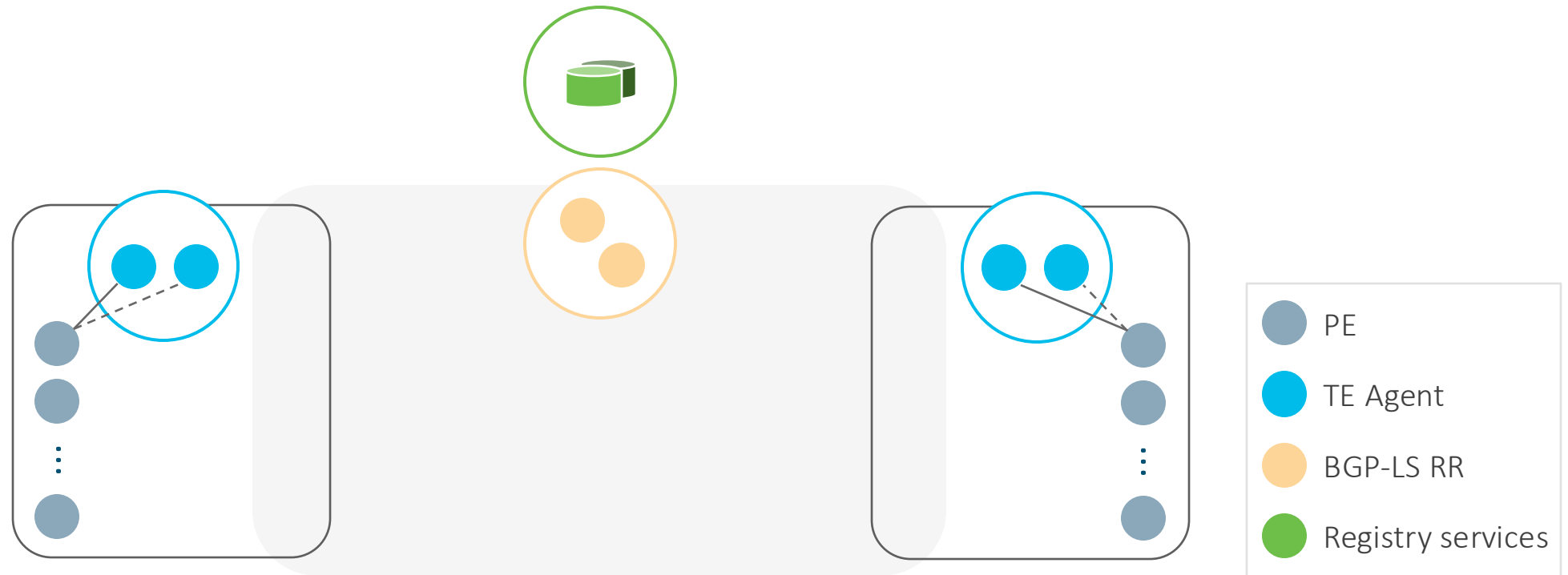
- Each headend connects to two TE Agents (or more)
- Upon losing connectivity to its primary TE Agent
 - Headend **replays** all policy requests to its backup and includes their **current paths**
 - Backup TE Agent **validates** the current paths, recomputing only those that are no longer valid

This prevents any gratuitous forwarding change
- FRR keeps ensuring **service continuity** in case of a network event during the switchover



Selective Coordination

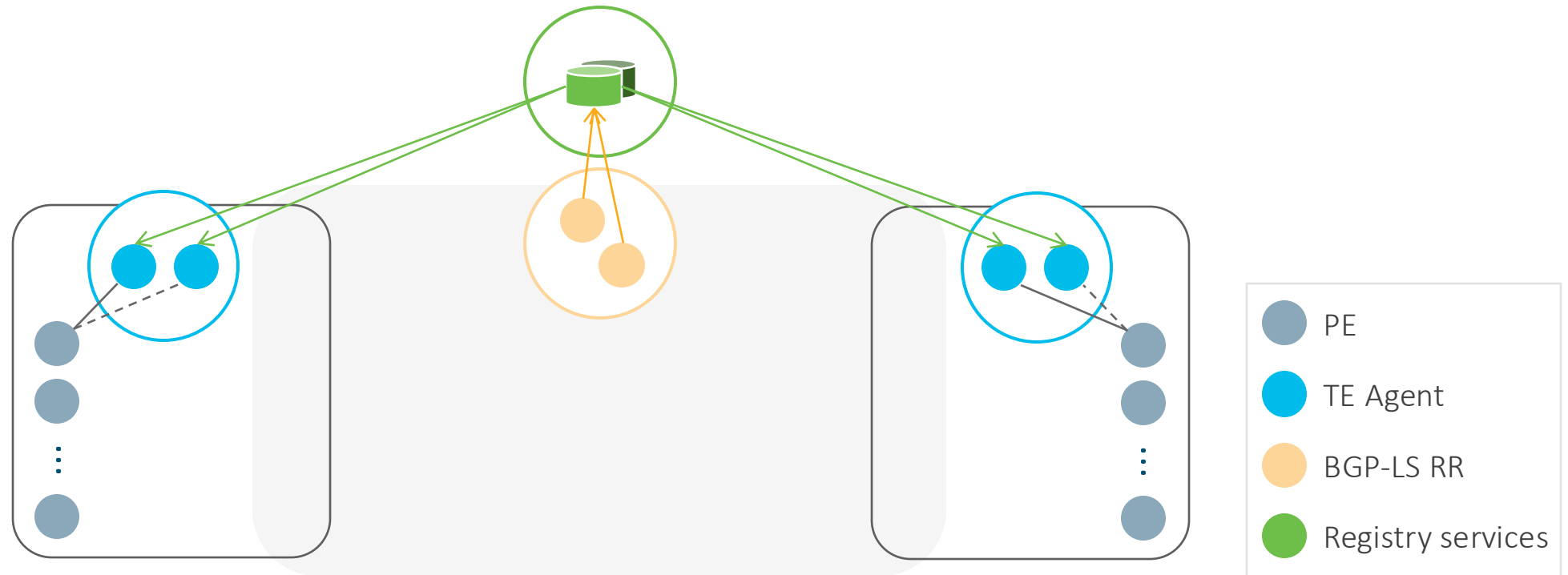
Selective Coordination



- Leverage **registry services** to coordinate shared resource utilization
 - **Bandwidth registry** serves as bookkeeper for bandwidth reservations
 - **Group registry** assigns a TE Agent for each disjoint (or bidirectional) group
- Based on time-proven, high-performance open-source database

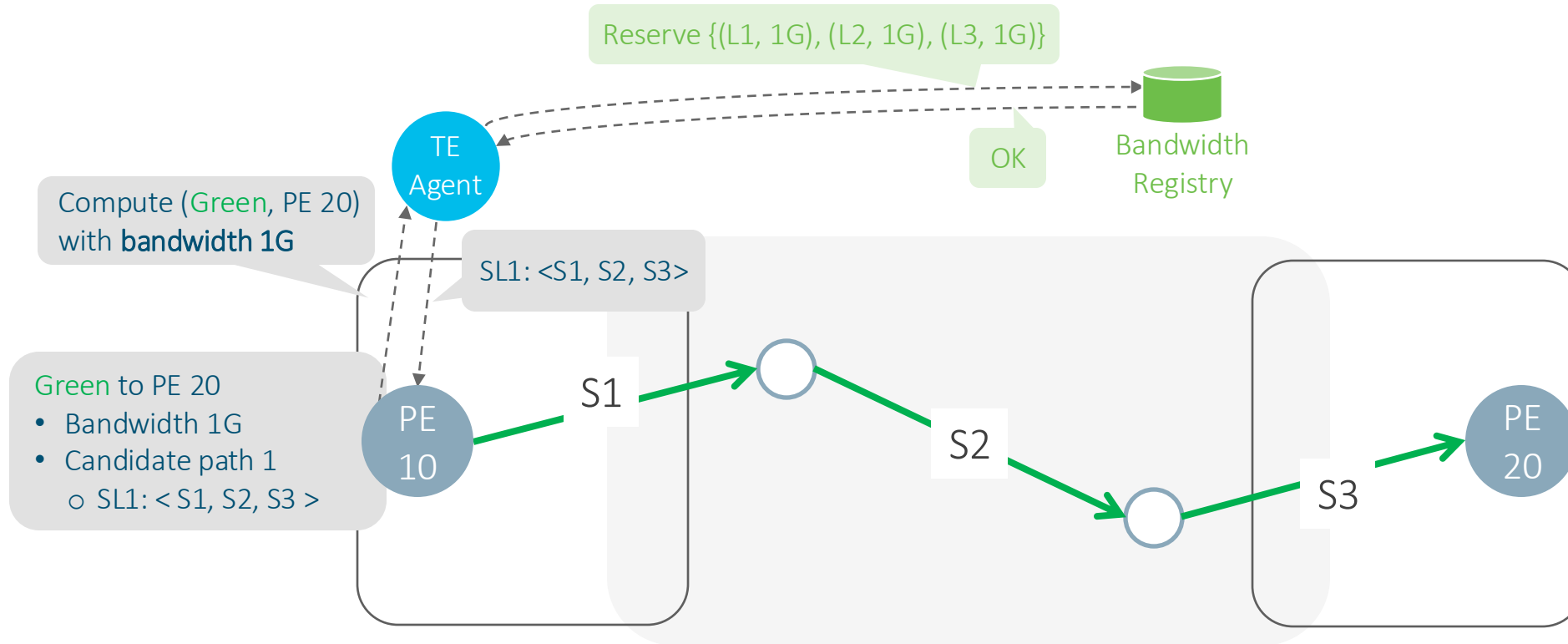
Bandwidth Reservation

Reservable Bandwidth Distribution



- Bandwidth Registry
 - Learns max reservable link bandwidth from BGP-LS RRs
 - Maintains a ledger of active reservations
 - Propagates available bandwidth to every TE Agent

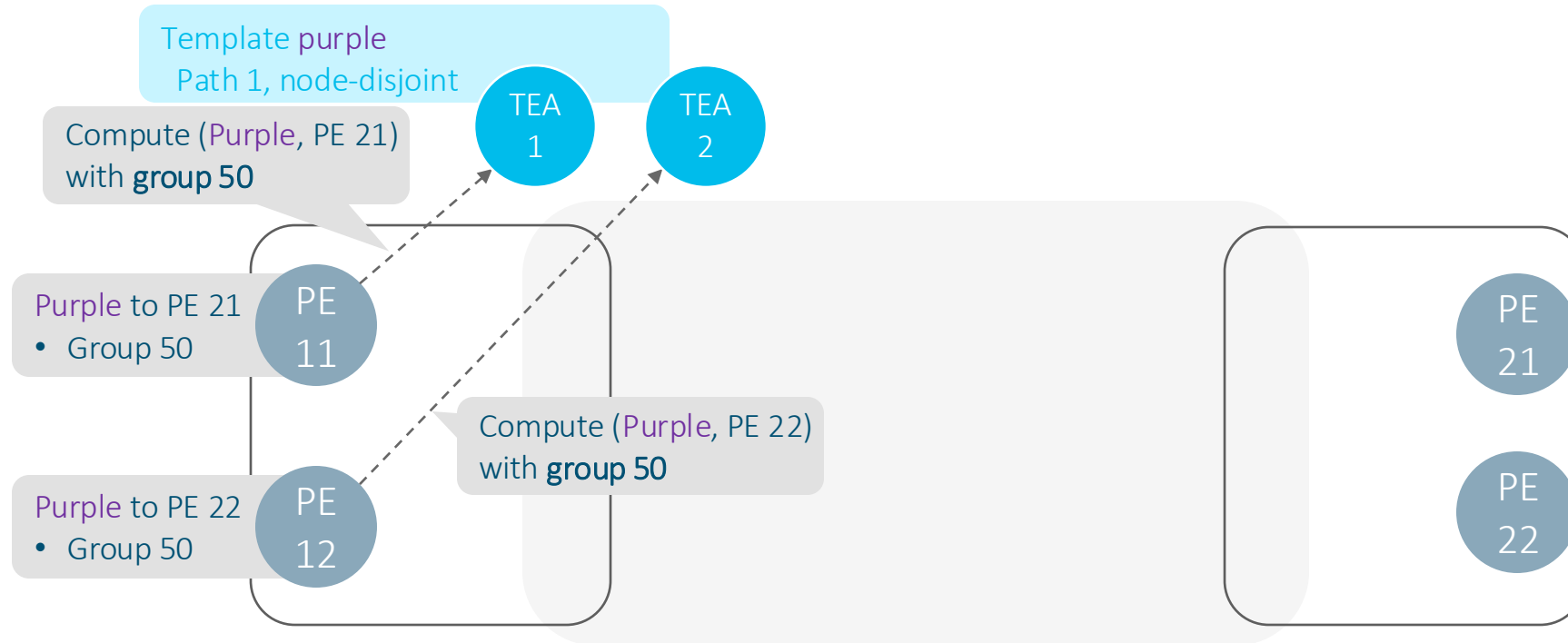
SR Policy Calculation with Bandwidth Reservation



- Ingress PE sends a compute request with a **bandwidth constraint**
- TE Agent computes strict, non-ECMP segment lists using Adj-SIDs
- TE Agent sends a **reservation request** to Bandwidth Registry
- Bandwidth Registry performs a **vector lock** for all links of the SR policy

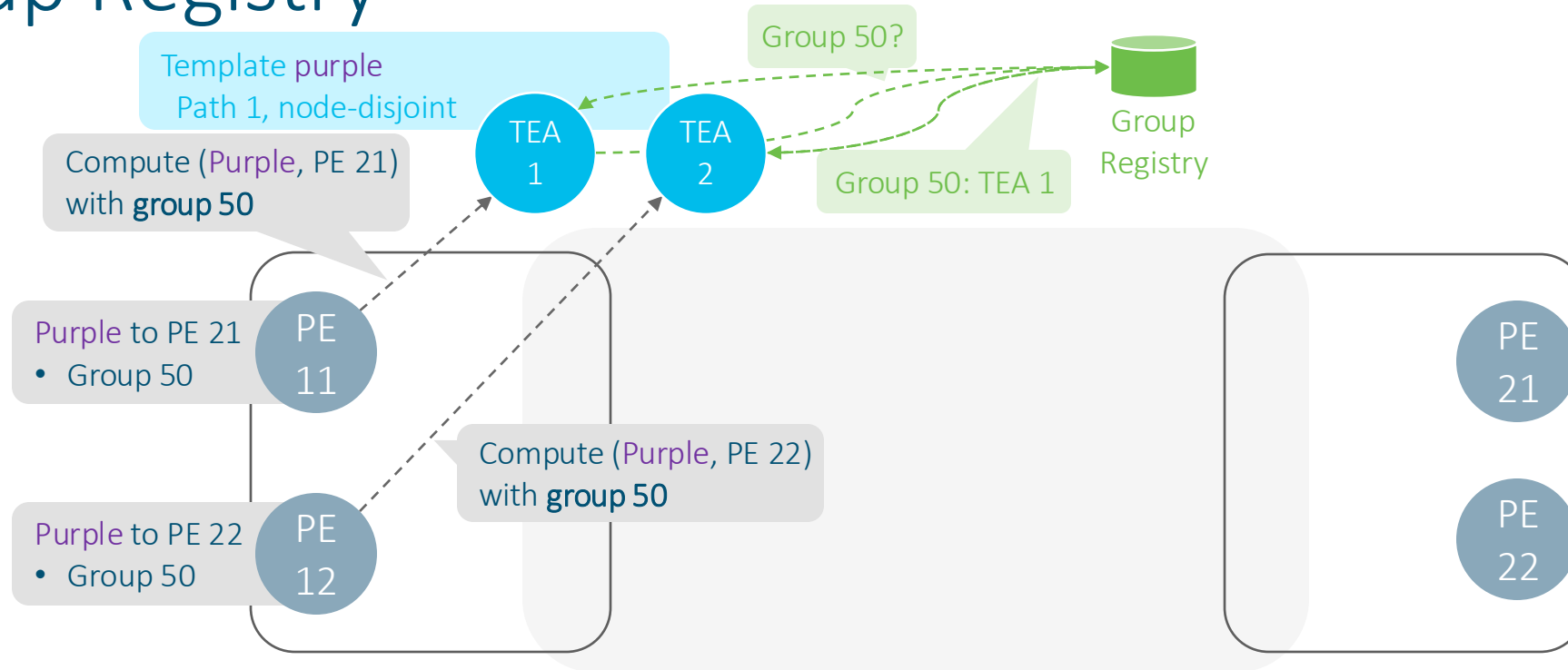
Disjoint Paths

Disjoint Pseudowires



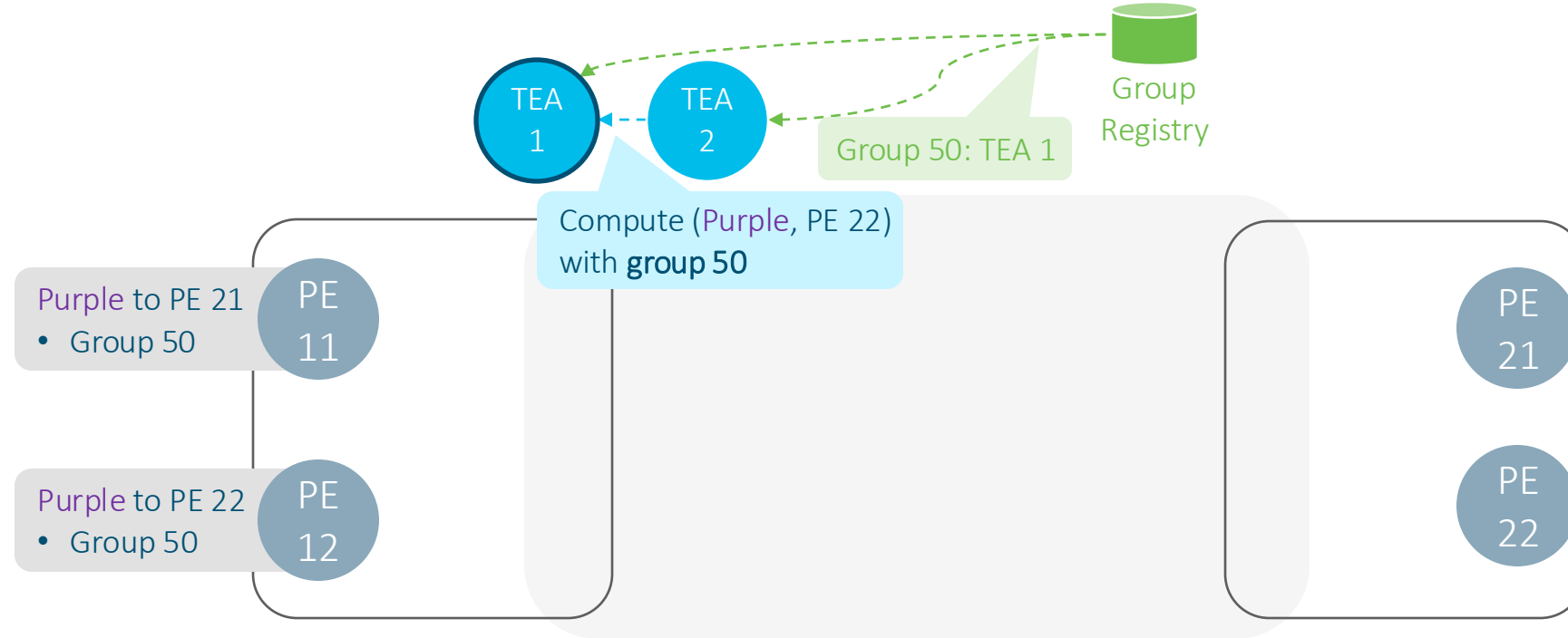
- Disjoint pseudowires with different headends and endpoints
- Each PE sends its compute request to its preferred TE agent
- How do we ensure disjoint paths?

Group Registry



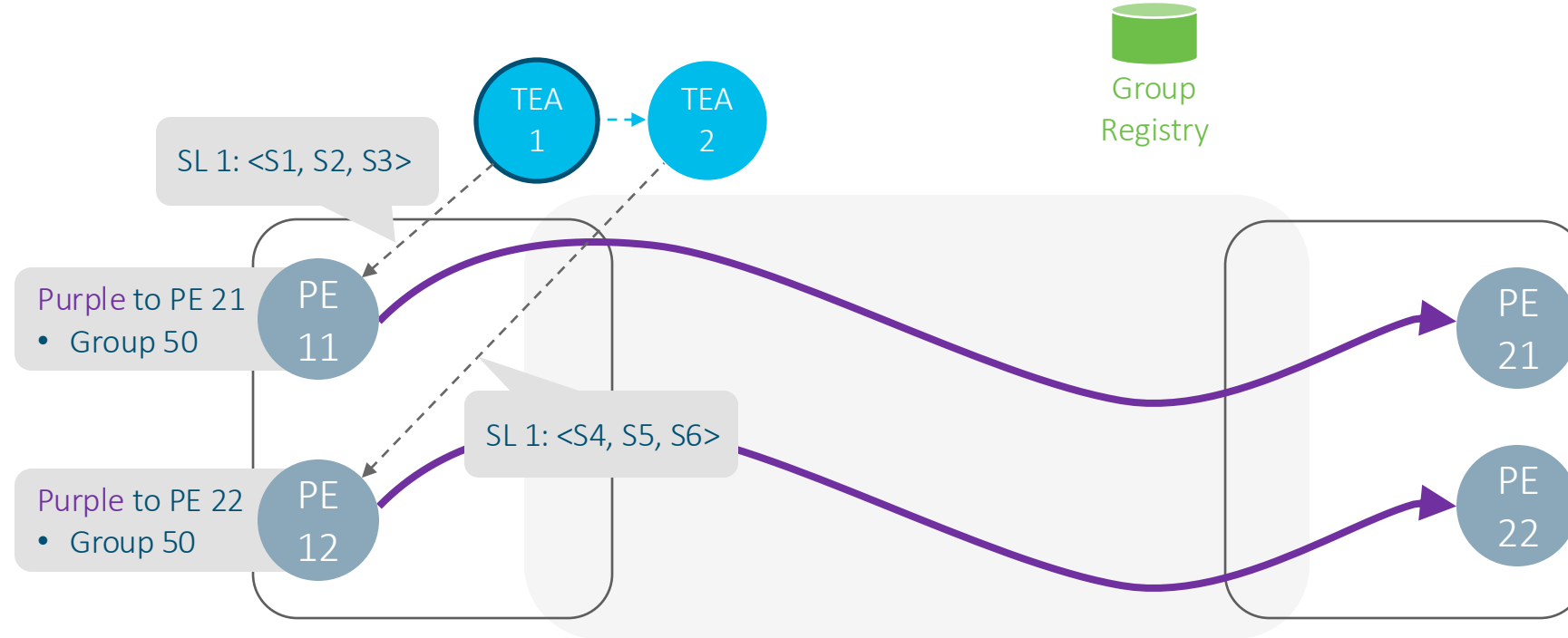
- Group Registry assigns a **designated TE Agent** for each group
 - The designated TE agent computes all policies in that group
 - Persistent storage of group assignments: <4MB for 100k groups

Relaying to Designated TE Agent



- Other TE agents relay the policy request to the **designated TE agent**
 - Relaying is invisible to the headend

Relaying to Designated TE Agent

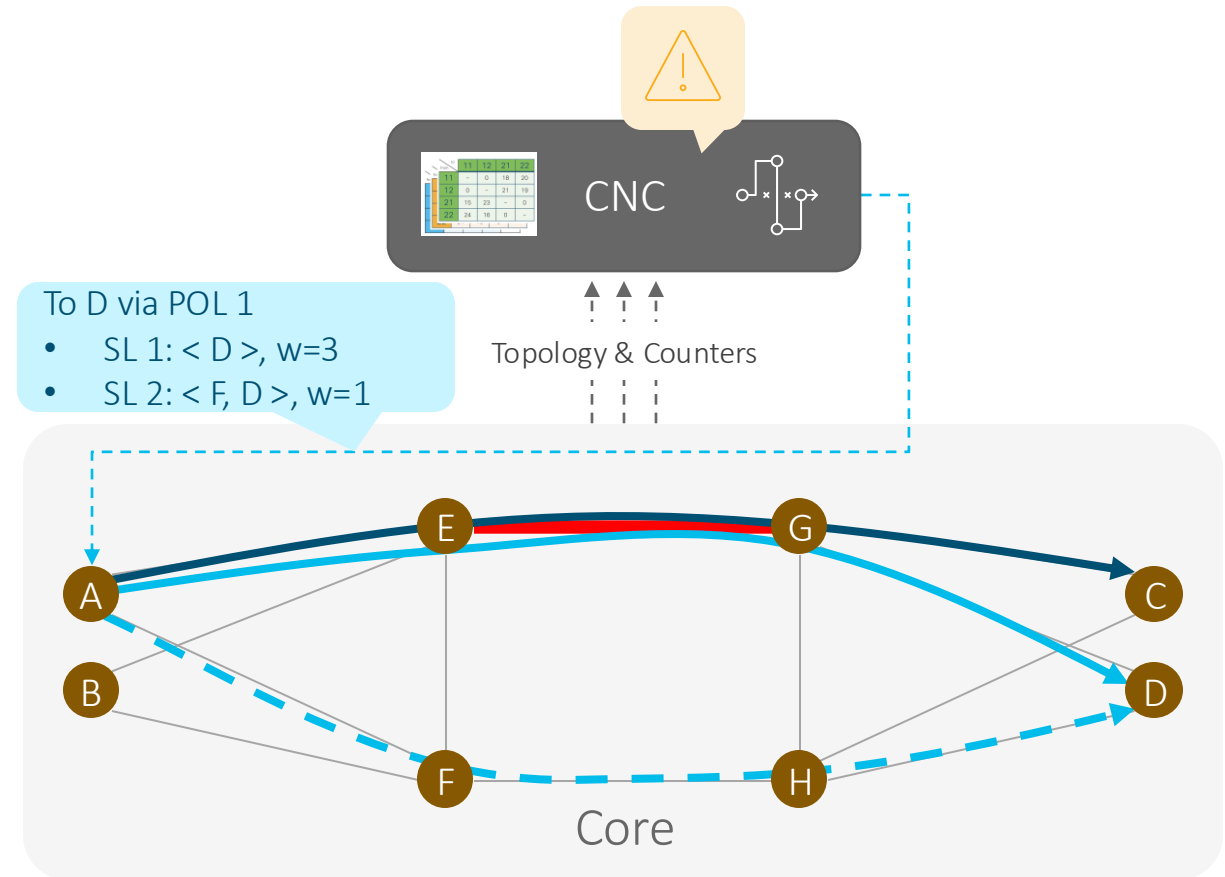


- Other TE agents relay the policy request to the **designated TE agent**
 - Relaying is invisible to the headend
- Response is relayed in the same way
- Same concept applies to co-routed bidirectional policies

Tactical Bandwidth Optimization

Tactical Bandwidth Optimization — Prepare for the Unplanned

- Detects imminent congestions
- Leverages **demand awareness** to compute the **least disruptive** alternate paths
 - Least amount of rerouted traffic
 - Lowest extra latency
- Provisions **tactical (ephemeral) SRv6 policies** and steer impacted flows

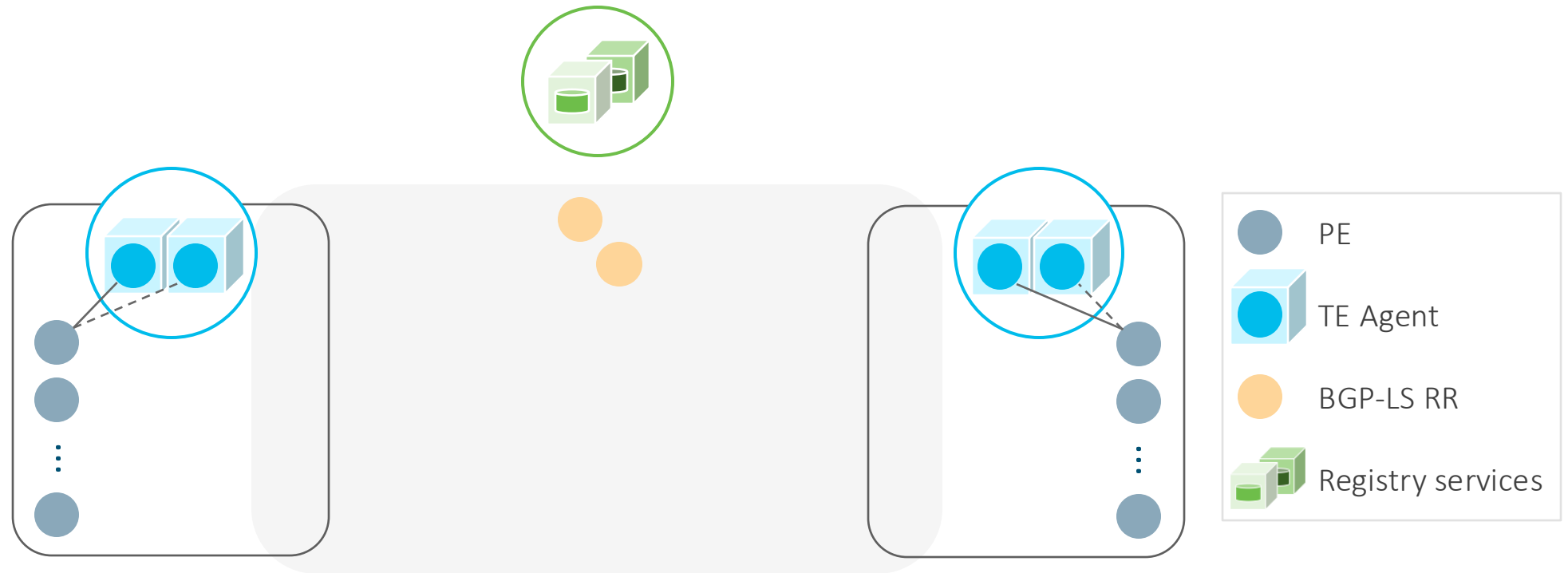


Allows the network to run *hotter* by tactically deviating traffic away from congestions.

Not an alternative to capacity planning!

Packaging & Deployment

DSDN Packaging & Deployment



- **TE Agents** are deployed as standalone VMs (CDG-like)
- **GW Agent** is provided as an XR capability (e.g., on XRv9k or any physical XR router)
- **Registry services** are provided as CNC capabilities
 - Leverage existing CNC databases and HA infrastructure (incl. geo-HA)

Roadmap Update

What's Next – SRv6 DSDN

1H CY26

IOS XR 26.2.1

- IP Measurement (IPM)

CNC 7.2

- Deterministic Demand Matrix (DDM) — Phase 1
- Automated Capacity Planning (ACP) — *Preview*

2H CY26

XR 26.4.1

- GW Agent (PCEP translation)

CNC 8.0

- TE Agent — Phase 1
Any metric, excl./incl. affinity, reserve bandwidth
- Bandwidth Registry
- DDM — Phase 2
Multi-domain, SR policies, better scale and accuracy

1H CY27

CNC 8.1

- ACP FCS
- Demand matrix export to Crosswork Planning

Radar

- TE Agent improvements
Disjoint paths, exclude IP, VIA, ..
- Tactical traffic engineering

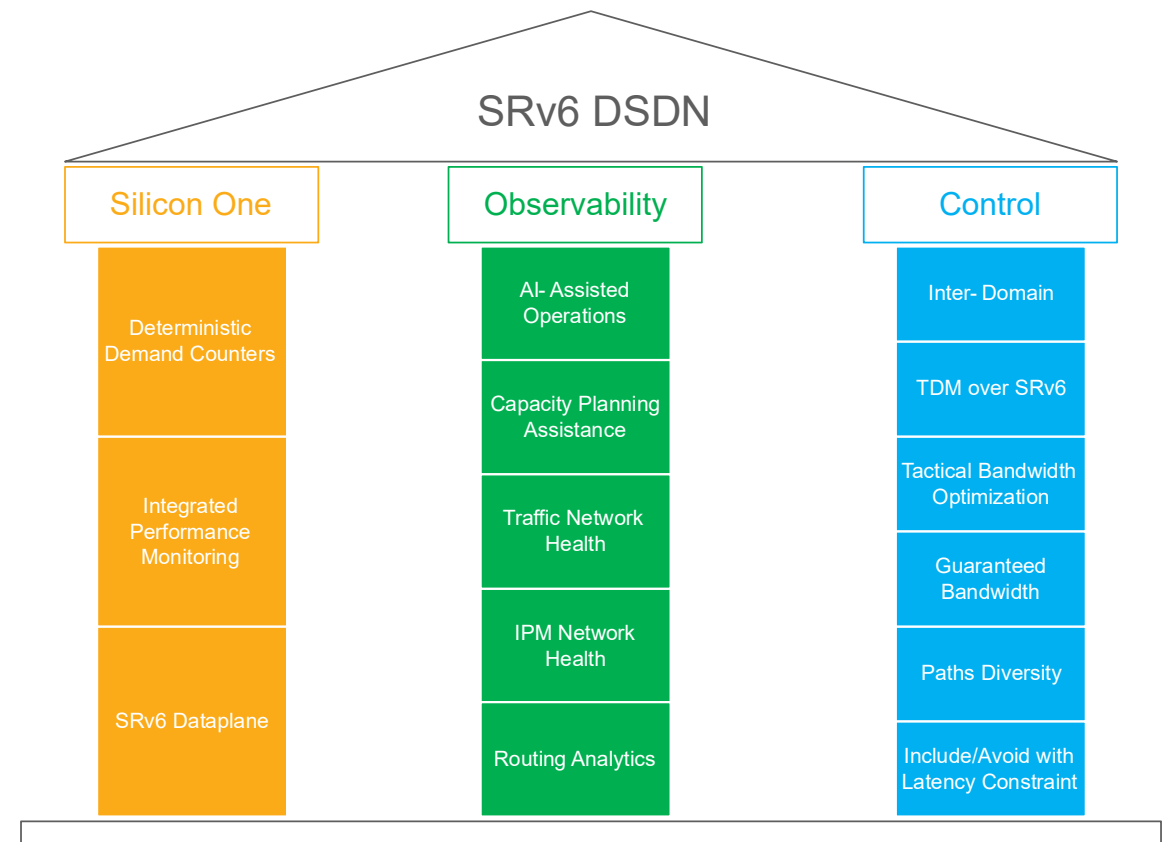
Some of the features described herein remain in varying stages of development and will be offered on a when-and-if-available basis. This roadmap is subject to change at the sole discretion of Cisco, and Cisco will have no liability for delay in the delivery or failure to deliver any of the products or features set forth in this presentation.

Simplicity Always Prevails

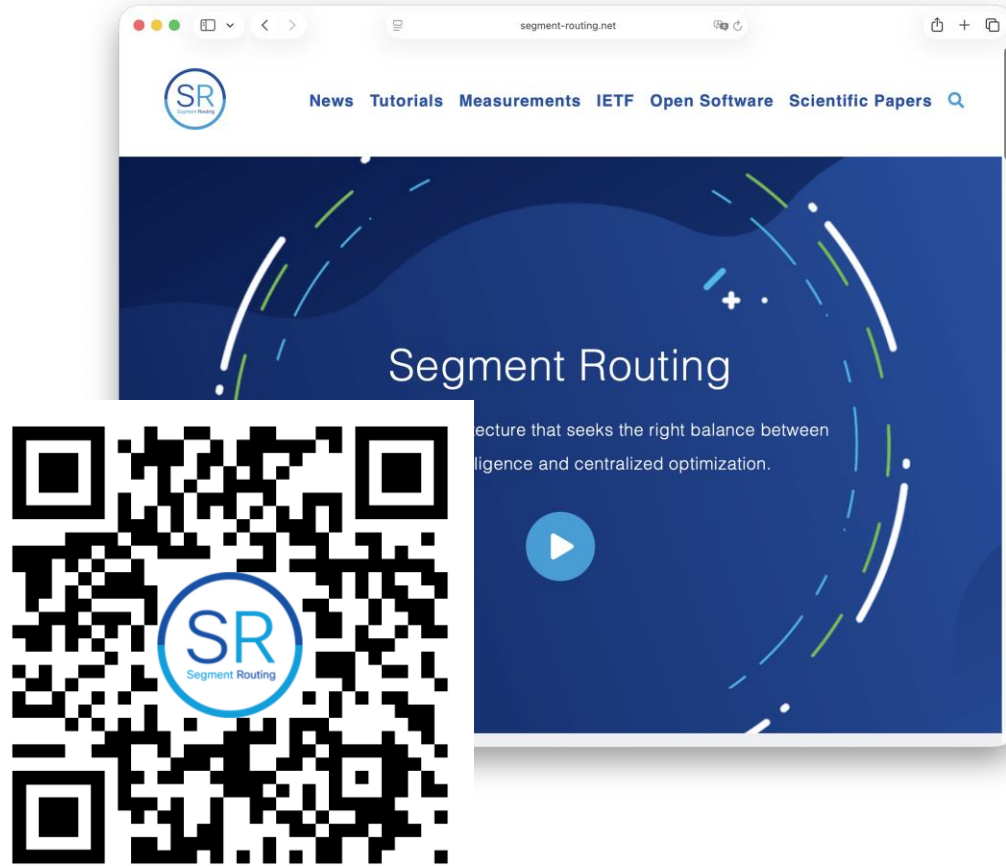


Three-Pillar Architecture

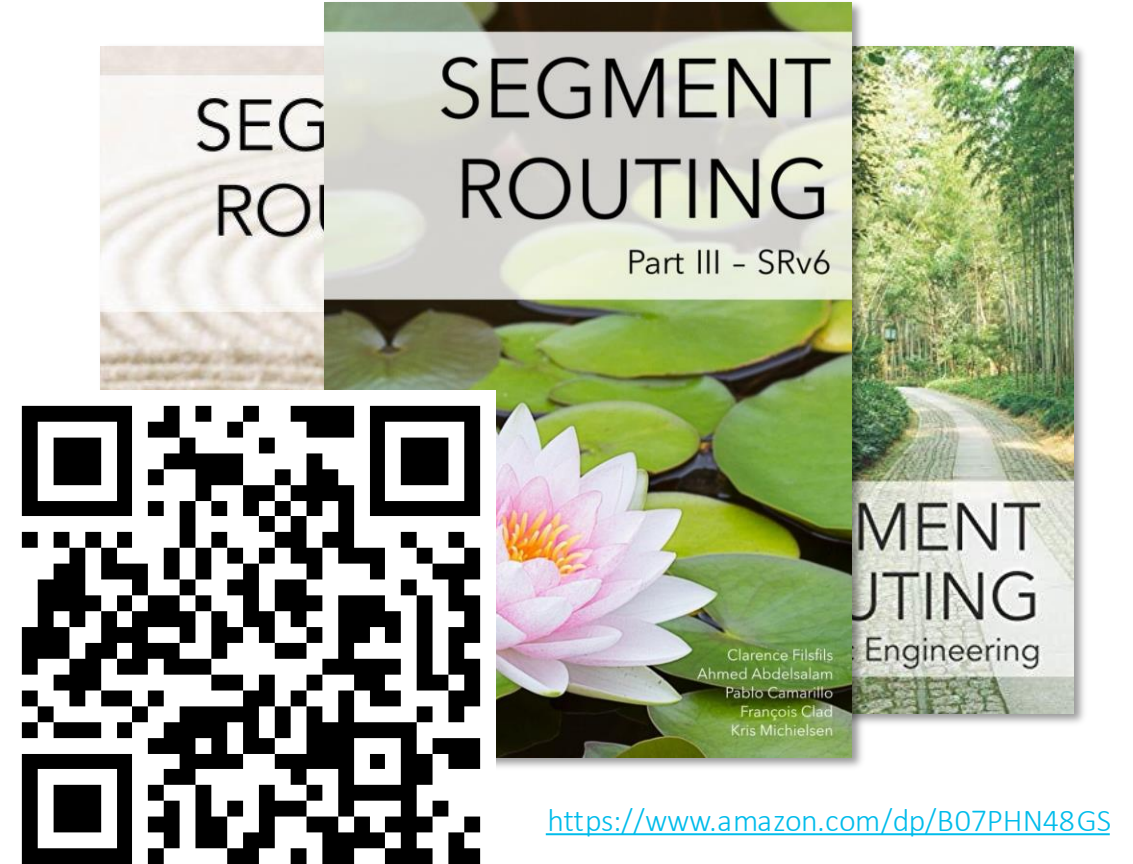
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